

**LABORATORIO DE ENSAYO DE MATERIALES**  
**INFORME DE ENSAYO N°096-25 SU22**

**CLIENTE** : LUZ DEL SUR  
**DIRECCIÓN \*\*** : AV. CANAVAL Y MOREYRA NRO. 380 LIMA - LIMA - SAN ISIDRO  
**PROYECTO \*\*** : EMS PARA CAMBIO DE POSTES-LINEAS AÉREAS  
**UBICACIÓN \*\*** : AV. PANAMERICANA SUR, DIST. SANTIAGO DE SURCO, LIMA, LIMA.

**CÓDIGO** : F-LEM-P-SU-22.02  
**RECEPCIÓN N°** : 267-25  
**F.EMISIÓN** : 2025-03-13

\*\* Datos proporcionados por el cliente

| <b>DATOS DE LA MUESTRA</b>                                                                                                                                                                                                                                                                                               |                                            |                                  |       |                                      |            |                      |           |                       |                                            |      |       |      |      |       |       |      |         |       |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                             |  |  |  |           |  |  |  |           |  |  |  |         |      |        |      |      |      |         |      |        |     |       |      |      |      |        |     |  |     |    |  |    |  |    |  |    |  |    |  |    |  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|----------------------------------|-------|--------------------------------------|------------|----------------------|-----------|-----------------------|--------------------------------------------|------|-------|------|------|-------|-------|------|---------|-------|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|--|--|--|-----------|--|--|--|-----------|--|--|--|---------|------|--------|------|------|------|---------|------|--------|-----|-------|------|------|------|--------|-----|--|-----|----|--|----|--|----|--|----|--|----|--|----|--|
| CANTERA/SONDAJE** : C1 / C-1 ESTRUCTURA E36LÍNEA L2013                                                                                                                                                                                                                                                                   |                                            | CÓDIGO DE LA MUESTRA : 423-SU-25 |       |                                      |            |                      |           |                       |                                            |      |       |      |      |       |       |      |         |       |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                             |  |  |  |           |  |  |  |           |  |  |  |         |      |        |      |      |      |         |      |        |     |       |      |      |      |        |     |  |     |    |  |    |  |    |  |    |  |    |  |    |  |
| N° MUESTRA ** : M-2                                                                                                                                                                                                                                                                                                      |                                            | FECHA DE RECEPCIÓN : 2025-03-07  |       |                                      |            |                      |           |                       |                                            |      |       |      |      |       |       |      |         |       |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                             |  |  |  |           |  |  |  |           |  |  |  |         |      |        |      |      |      |         |      |        |     |       |      |      |      |        |     |  |     |    |  |    |  |    |  |    |  |    |  |    |  |
| TIPO DE MUESTRA ** : SUELO                                                                                                                                                                                                                                                                                               |                                            | FECHA DE EJECUCIÓN : 2025-03-08  |       |                                      |            |                      |           |                       |                                            |      |       |      |      |       |       |      |         |       |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                             |  |  |  |           |  |  |  |           |  |  |  |         |      |        |      |      |      |         |      |        |     |       |      |      |      |        |     |  |     |    |  |    |  |    |  |    |  |    |  |    |  |
| LUGAR DE ENSAYO : LABORATORIO DE ENSAYOS DE MATERIALES                                                                                                                                                                                                                                                                   |                                            |                                  |       |                                      |            |                      |           |                       |                                            |      |       |      |      |       |       |      |         |       |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                             |  |  |  |           |  |  |  |           |  |  |  |         |      |        |      |      |      |         |      |        |     |       |      |      |      |        |     |  |     |    |  |    |  |    |  |    |  |    |  |    |  |
| <table><tr><th colspan="2">Tamiz</th><th rowspan="2">% que Pasa</th></tr><tr><th>in.</th><th>mm.</th></tr><tr><td>No.4</td><td>4.75</td><td>40.7</td></tr><tr><td>No.10</td><td>2.00</td><td>35.5</td></tr><tr><td>No.40</td><td>0.425</td><td>24.6</td></tr><tr><td>No. 200</td><td>0.075</td><td>5.5</td></tr></table> |                                            |                                  | Tamiz |                                      | % que Pasa | in.                  | mm.       | No.4                  | 4.75                                       | 40.7 | No.10 | 2.00 | 35.5 | No.40 | 0.425 | 24.6 | No. 200 | 0.075 | 5.5 | <table><tr><th colspan="4">Distribución granulometrica</th></tr><tr><td colspan="2">% BOLONES</td><td colspan="2"></td></tr><tr><td colspan="2">% BLOQUES</td><td colspan="2"></td></tr><tr><td rowspan="2">% GRAVA</td><td rowspan="2">59.3</td><td>Gruesa</td><td>37.5</td></tr><tr><td>Fina</td><td>21.9</td></tr><tr><td rowspan="3">% ARENA</td><td rowspan="3">35.2</td><td>Gruesa</td><td>5.1</td></tr><tr><td>Media</td><td>18.6</td></tr><tr><td>Fina</td><td>11.4</td></tr><tr><td>% FINO</td><td>5.5</td><td></td><td>5.5</td></tr><tr><td colspan="2">LL</td><td colspan="2">NP</td></tr><tr><td colspan="2">LP</td><td colspan="2">NP</td></tr><tr><td colspan="2">IP</td><td colspan="2">NP</td></tr></table> | Distribución granulometrica |  |  |  | % BOLONES |  |  |  | % BLOQUES |  |  |  | % GRAVA | 59.3 | Gruesa | 37.5 | Fina | 21.9 | % ARENA | 35.2 | Gruesa | 5.1 | Media | 18.6 | Fina | 11.4 | % FINO | 5.5 |  | 5.5 | LL |  | NP |  | LP |  | NP |  | IP |  | NP |  |
| Tamiz                                                                                                                                                                                                                                                                                                                    |                                            | % que Pasa                       |       |                                      |            |                      |           |                       |                                            |      |       |      |      |       |       |      |         |       |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                             |  |  |  |           |  |  |  |           |  |  |  |         |      |        |      |      |      |         |      |        |     |       |      |      |      |        |     |  |     |    |  |    |  |    |  |    |  |    |  |    |  |
| in.                                                                                                                                                                                                                                                                                                                      | mm.                                        |                                  |       |                                      |            |                      |           |                       |                                            |      |       |      |      |       |       |      |         |       |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                             |  |  |  |           |  |  |  |           |  |  |  |         |      |        |      |      |      |         |      |        |     |       |      |      |      |        |     |  |     |    |  |    |  |    |  |    |  |    |  |    |  |
| No.4                                                                                                                                                                                                                                                                                                                     | 4.75                                       | 40.7                             |       |                                      |            |                      |           |                       |                                            |      |       |      |      |       |       |      |         |       |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                             |  |  |  |           |  |  |  |           |  |  |  |         |      |        |      |      |      |         |      |        |     |       |      |      |      |        |     |  |     |    |  |    |  |    |  |    |  |    |  |    |  |
| No.10                                                                                                                                                                                                                                                                                                                    | 2.00                                       | 35.5                             |       |                                      |            |                      |           |                       |                                            |      |       |      |      |       |       |      |         |       |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                             |  |  |  |           |  |  |  |           |  |  |  |         |      |        |      |      |      |         |      |        |     |       |      |      |      |        |     |  |     |    |  |    |  |    |  |    |  |    |  |    |  |
| No.40                                                                                                                                                                                                                                                                                                                    | 0.425                                      | 24.6                             |       |                                      |            |                      |           |                       |                                            |      |       |      |      |       |       |      |         |       |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                             |  |  |  |           |  |  |  |           |  |  |  |         |      |        |      |      |      |         |      |        |     |       |      |      |      |        |     |  |     |    |  |    |  |    |  |    |  |    |  |    |  |
| No. 200                                                                                                                                                                                                                                                                                                                  | 0.075                                      | 5.5                              |       |                                      |            |                      |           |                       |                                            |      |       |      |      |       |       |      |         |       |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                             |  |  |  |           |  |  |  |           |  |  |  |         |      |        |      |      |      |         |      |        |     |       |      |      |      |        |     |  |     |    |  |    |  |    |  |    |  |    |  |    |  |
| Distribución granulometrica                                                                                                                                                                                                                                                                                              |                                            |                                  |       |                                      |            |                      |           |                       |                                            |      |       |      |      |       |       |      |         |       |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                             |  |  |  |           |  |  |  |           |  |  |  |         |      |        |      |      |      |         |      |        |     |       |      |      |      |        |     |  |     |    |  |    |  |    |  |    |  |    |  |    |  |
| % BOLONES                                                                                                                                                                                                                                                                                                                |                                            |                                  |       |                                      |            |                      |           |                       |                                            |      |       |      |      |       |       |      |         |       |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                             |  |  |  |           |  |  |  |           |  |  |  |         |      |        |      |      |      |         |      |        |     |       |      |      |      |        |     |  |     |    |  |    |  |    |  |    |  |    |  |    |  |
| % BLOQUES                                                                                                                                                                                                                                                                                                                |                                            |                                  |       |                                      |            |                      |           |                       |                                            |      |       |      |      |       |       |      |         |       |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                             |  |  |  |           |  |  |  |           |  |  |  |         |      |        |      |      |      |         |      |        |     |       |      |      |      |        |     |  |     |    |  |    |  |    |  |    |  |    |  |    |  |
| % GRAVA                                                                                                                                                                                                                                                                                                                  | 59.3                                       | Gruesa                           | 37.5  |                                      |            |                      |           |                       |                                            |      |       |      |      |       |       |      |         |       |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                             |  |  |  |           |  |  |  |           |  |  |  |         |      |        |      |      |      |         |      |        |     |       |      |      |      |        |     |  |     |    |  |    |  |    |  |    |  |    |  |    |  |
|                                                                                                                                                                                                                                                                                                                          |                                            | Fina                             | 21.9  |                                      |            |                      |           |                       |                                            |      |       |      |      |       |       |      |         |       |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                             |  |  |  |           |  |  |  |           |  |  |  |         |      |        |      |      |      |         |      |        |     |       |      |      |      |        |     |  |     |    |  |    |  |    |  |    |  |    |  |    |  |
| % ARENA                                                                                                                                                                                                                                                                                                                  | 35.2                                       | Gruesa                           | 5.1   |                                      |            |                      |           |                       |                                            |      |       |      |      |       |       |      |         |       |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                             |  |  |  |           |  |  |  |           |  |  |  |         |      |        |      |      |      |         |      |        |     |       |      |      |      |        |     |  |     |    |  |    |  |    |  |    |  |    |  |    |  |
|                                                                                                                                                                                                                                                                                                                          |                                            | Media                            | 18.6  |                                      |            |                      |           |                       |                                            |      |       |      |      |       |       |      |         |       |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                             |  |  |  |           |  |  |  |           |  |  |  |         |      |        |      |      |      |         |      |        |     |       |      |      |      |        |     |  |     |    |  |    |  |    |  |    |  |    |  |    |  |
|                                                                                                                                                                                                                                                                                                                          |                                            | Fina                             | 11.4  |                                      |            |                      |           |                       |                                            |      |       |      |      |       |       |      |         |       |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                             |  |  |  |           |  |  |  |           |  |  |  |         |      |        |      |      |      |         |      |        |     |       |      |      |      |        |     |  |     |    |  |    |  |    |  |    |  |    |  |    |  |
| % FINO                                                                                                                                                                                                                                                                                                                   | 5.5                                        |                                  | 5.5   |                                      |            |                      |           |                       |                                            |      |       |      |      |       |       |      |         |       |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                             |  |  |  |           |  |  |  |           |  |  |  |         |      |        |      |      |      |         |      |        |     |       |      |      |      |        |     |  |     |    |  |    |  |    |  |    |  |    |  |    |  |
| LL                                                                                                                                                                                                                                                                                                                       |                                            | NP                               |       |                                      |            |                      |           |                       |                                            |      |       |      |      |       |       |      |         |       |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                             |  |  |  |           |  |  |  |           |  |  |  |         |      |        |      |      |      |         |      |        |     |       |      |      |      |        |     |  |     |    |  |    |  |    |  |    |  |    |  |    |  |
| LP                                                                                                                                                                                                                                                                                                                       |                                            | NP                               |       |                                      |            |                      |           |                       |                                            |      |       |      |      |       |       |      |         |       |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                             |  |  |  |           |  |  |  |           |  |  |  |         |      |        |      |      |      |         |      |        |     |       |      |      |      |        |     |  |     |    |  |    |  |    |  |    |  |    |  |    |  |
| IP                                                                                                                                                                                                                                                                                                                       |                                            | NP                               |       |                                      |            |                      |           |                       |                                            |      |       |      |      |       |       |      |         |       |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                             |  |  |  |           |  |  |  |           |  |  |  |         |      |        |      |      |      |         |      |        |     |       |      |      |      |        |     |  |     |    |  |    |  |    |  |    |  |    |  |    |  |
| <table><tr><td>D10</td><td>0.151</td></tr><tr><td>D30</td><td>0.8</td></tr><tr><td>D60</td><td>17.2</td></tr><tr><td>Cu</td><td>114.2</td></tr><tr><td>Cc</td><td>0.2</td></tr></table>                                                                                                                                  |                                            |                                  |       | D10                                  | 0.151      | D30                  | 0.8       | D60                   | 17.2                                       | Cu   | 114.2 | Cc   | 0.2  |       |       |      |         |       |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                             |  |  |  |           |  |  |  |           |  |  |  |         |      |        |      |      |      |         |      |        |     |       |      |      |      |        |     |  |     |    |  |    |  |    |  |    |  |    |  |    |  |
| D10                                                                                                                                                                                                                                                                                                                      | 0.151                                      |                                  |       |                                      |            |                      |           |                       |                                            |      |       |      |      |       |       |      |         |       |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                             |  |  |  |           |  |  |  |           |  |  |  |         |      |        |      |      |      |         |      |        |     |       |      |      |      |        |     |  |     |    |  |    |  |    |  |    |  |    |  |    |  |
| D30                                                                                                                                                                                                                                                                                                                      | 0.8                                        |                                  |       |                                      |            |                      |           |                       |                                            |      |       |      |      |       |       |      |         |       |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                             |  |  |  |           |  |  |  |           |  |  |  |         |      |        |      |      |      |         |      |        |     |       |      |      |      |        |     |  |     |    |  |    |  |    |  |    |  |    |  |    |  |
| D60                                                                                                                                                                                                                                                                                                                      | 17.2                                       |                                  |       |                                      |            |                      |           |                       |                                            |      |       |      |      |       |       |      |         |       |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                             |  |  |  |           |  |  |  |           |  |  |  |         |      |        |      |      |      |         |      |        |     |       |      |      |      |        |     |  |     |    |  |    |  |    |  |    |  |    |  |    |  |
| Cu                                                                                                                                                                                                                                                                                                                       | 114.2                                      |                                  |       |                                      |            |                      |           |                       |                                            |      |       |      |      |       |       |      |         |       |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                             |  |  |  |           |  |  |  |           |  |  |  |         |      |        |      |      |      |         |      |        |     |       |      |      |      |        |     |  |     |    |  |    |  |    |  |    |  |    |  |    |  |
| Cc                                                                                                                                                                                                                                                                                                                       | 0.2                                        |                                  |       |                                      |            |                      |           |                       |                                            |      |       |      |      |       |       |      |         |       |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                             |  |  |  |           |  |  |  |           |  |  |  |         |      |        |      |      |      |         |      |        |     |       |      |      |      |        |     |  |     |    |  |    |  |    |  |    |  |    |  |    |  |
| <b>Standard Practice for Classification of Soils for Engineering Purposes (Unified Soil Classification System)<sup>1</sup></b><br><b>D2487 - 17<sup>E1</sup></b>                                                                                                                                                         |                                            |                                  |       |                                      |            |                      |           |                       |                                            |      |       |      |      |       |       |      |         |       |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                             |  |  |  |           |  |  |  |           |  |  |  |         |      |        |      |      |      |         |      |        |     |       |      |      |      |        |     |  |     |    |  |    |  |    |  |    |  |    |  |    |  |
| <table><tr><th colspan="2">SISTEMA UNIFICADO CLASIFICACIÓN SUCS</th></tr><tr><td>Simbolo de Grupo</td><td>GP-GM</td></tr><tr><td>Denominación de Grupo</td><td>Grava pobremente graduada con limo y arena</td></tr></table>                                                                                              |                                            |                                  |       | SISTEMA UNIFICADO CLASIFICACIÓN SUCS |            | Simbolo de Grupo     | GP-GM     | Denominación de Grupo | Grava pobremente graduada con limo y arena |      |       |      |      |       |       |      |         |       |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                             |  |  |  |           |  |  |  |           |  |  |  |         |      |        |      |      |      |         |      |        |     |       |      |      |      |        |     |  |     |    |  |    |  |    |  |    |  |    |  |    |  |
| SISTEMA UNIFICADO CLASIFICACIÓN SUCS                                                                                                                                                                                                                                                                                     |                                            |                                  |       |                                      |            |                      |           |                       |                                            |      |       |      |      |       |       |      |         |       |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                             |  |  |  |           |  |  |  |           |  |  |  |         |      |        |      |      |      |         |      |        |     |       |      |      |      |        |     |  |     |    |  |    |  |    |  |    |  |    |  |    |  |
| Simbolo de Grupo                                                                                                                                                                                                                                                                                                         | GP-GM                                      |                                  |       |                                      |            |                      |           |                       |                                            |      |       |      |      |       |       |      |         |       |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                             |  |  |  |           |  |  |  |           |  |  |  |         |      |        |      |      |      |         |      |        |     |       |      |      |      |        |     |  |     |    |  |    |  |    |  |    |  |    |  |    |  |
| Denominación de Grupo                                                                                                                                                                                                                                                                                                    | Grava pobremente graduada con limo y arena |                                  |       |                                      |            |                      |           |                       |                                            |      |       |      |      |       |       |      |         |       |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                             |  |  |  |           |  |  |  |           |  |  |  |         |      |        |      |      |      |         |      |        |     |       |      |      |      |        |     |  |     |    |  |    |  |    |  |    |  |    |  |    |  |
| <b>Standard Practice for Classification of Soils and Soil-Aggregate Mixtures for Highway Construction Purposes<sup>1</sup></b><br><b>D3282-24</b>                                                                                                                                                                        |                                            |                                  |       |                                      |            |                      |           |                       |                                            |      |       |      |      |       |       |      |         |       |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                             |  |  |  |           |  |  |  |           |  |  |  |         |      |        |      |      |      |         |      |        |     |       |      |      |      |        |     |  |     |    |  |    |  |    |  |    |  |    |  |    |  |
| <table><tr><th colspan="2">SISTEMA DE CLASIFICACION AASHTO</th></tr><tr><td>Clasificación AASHTO</td><td>A-1-a (0)</td></tr></table>                                                                                                                                                                                     |                                            |                                  |       | SISTEMA DE CLASIFICACION AASHTO      |            | Clasificación AASHTO | A-1-a (0) |                       |                                            |      |       |      |      |       |       |      |         |       |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                             |  |  |  |           |  |  |  |           |  |  |  |         |      |        |      |      |      |         |      |        |     |       |      |      |      |        |     |  |     |    |  |    |  |    |  |    |  |    |  |    |  |
| SISTEMA DE CLASIFICACION AASHTO                                                                                                                                                                                                                                                                                          |                                            |                                  |       |                                      |            |                      |           |                       |                                            |      |       |      |      |       |       |      |         |       |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                             |  |  |  |           |  |  |  |           |  |  |  |         |      |        |      |      |      |         |      |        |     |       |      |      |      |        |     |  |     |    |  |    |  |    |  |    |  |    |  |    |  |
| Clasificación AASHTO                                                                                                                                                                                                                                                                                                     | A-1-a (0)                                  |                                  |       |                                      |            |                      |           |                       |                                            |      |       |      |      |       |       |      |         |       |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                             |  |  |  |           |  |  |  |           |  |  |  |         |      |        |      |      |      |         |      |        |     |       |      |      |      |        |     |  |     |    |  |    |  |    |  |    |  |    |  |    |  |

**Ensayos de referencia:**

La distribución granulometrica corresponde al Informe de ensayo N°105-25 SU24

El limite de Atterberg corresponde al Informe de ensayo N°119-25 SU23

Colocar en observaciones lo siguiente : Correspondiente al proyecto CAMBIO DE ESTRUCTURAS METÁLICAS E27, E35 Y E36 L2013 220 KV

  
IRMA COAQUIRA LAYME  
Ingeniero Civil CIP 121204  
Laboratorio Geofal S.A.C.

